



Day 12 – SwiftUI Tip Calculator App

Today I built a simple **Tip Calculator App** using SwiftUI and learned:

- `@State`
- Computed Properties
- `TextField`
- `Picker`
- Segmented Picker
- Currency Formatting
- `@FocusState`
- Toolbar keyboard handling
- Dynamic calculations in SwiftUI

What I Learned

- Managed dynamic UI state using `@State`
- Created computed properties for calculations
- Built forms using `Form` & `Section`
- Used `TextField` with currency formatting
- Used `Picker` for selecting people count
- Created segmented picker UI
- Controlled keyboard focus using `@FocusState`
- Added keyboard toolbar button using `.toolbar`

@State

`@State` stores values that can change and automatically updates the UI.

```
```swift
@State private var checkAmount = 0.0
@State private var numberOfpeople = 2
@State private var tipPercentage = 20
```

---

## Computed Property

Computed properties calculate values dynamically.

```
var totalPerperson: Double {

 let peopleCount = Double(numberOfpeople + 2)
 let tipSelection = Double(tipPercentage)

 let tipValue = checkAmount / 100 * tipSelection
 let grandTotal = checkAmount + tipValue

 return grandTotal / peopleCount
}
```

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## ◆ TextField with Currency Formatting

```
TextField(
 "Amount",
 value: $checkAmount,
 format: .currency(
 code: Locale.current.currency?.identifier ?? "INR"
)
)
.keyboardType(.decimalPad)
```

---

## ◆ Picker

Used for selecting values.

```
Picker("Kitna Ladka logo ko", selection: $numberOfpeople) {

 ForEach(2..<100) {
 Text("\($0) ladka log")
 }
}
```

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## ◆ Segmented Picker

```
Picker("TipPercentage", selection: $tipPercentage) {

 ForEach(tipPercentages, id: \.self) {
 Text("\($0)%")
 }
}
.pickerStyle(.segmented)
```

---

## ◆ @FocusState

Used to manage keyboard focus.

```
@FocusState private var amountIsFocused: Bool
```

Usage:

```
.focused($amountIsFocused)
```

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## ◆ Toolbar Keyboard Button

```
.toolbar {
 ToolbarItemGroup(placement: .keyboard) {
 Spacer()
 Button("Done") {
 amountIsFocused = false
 }
 }
}
```

This hides the keyboard when tapping **Done**.

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## ◆ Navigation Title

```
.navigationTitle("SwiftUI Learning 🚀")
```

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## 🎯 Important Interview Questions

### 1 What is `@State` in SwiftUI?

`@State` stores mutable local data and refreshes the UI automatically when values change.

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### 2 What is a Computed Property?

A property that calculates and returns a value instead of storing it.

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### 3 Why do we use `@FocusState`?

`@FocusState` controls keyboard focus and input handling.

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### 4 What is `.pickerStyle(.segmented)`?

It changes the Picker UI into a segmented control.

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## 5 What is Binding in SwiftUI?

Binding creates a two-way connection between UI and data.

Example:

```
$checkAmount
```

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## 6 Why do we use `.toolbar` in SwiftUI?

`.toolbar` is used to add buttons or controls to navigation bars or keyboards.

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## ★ Quick Revision

- `@State` = reactive UI state
  - Computed Property = calculated value
  - `TextField` = user input
  - `Picker` = selection control
  - `.segmented` = segmented picker UI
  - `@FocusState` = keyboard control
  - `.toolbar` = custom toolbar items
  - `$` = Binding
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## Day 12 Summary

Today I built a functional SwiftUI Tip Calculator app using:

- Forms
- Pickers
- Segmented controls
- Currency formatting
- Dynamic calculations
- Keyboard management
- Reactive UI updates